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HOUSE PRICE PREDICTION USING MULTIPLE LINEAR REGRESSION

1. Introduction and Objective

House prices are influenced by several factors, including the size of a property, its age, its location, and its distance from important urban areas. Because of these factors, machine learning can be used to identify relationships between property characteristics and house prices and then use those relationships to estimate the price of a new property.

2. Data Cleaning and Preparation

The original house-price dataset was inspected and prepared before training the machine learning model. Data preparation was necessary to ensure that the information supplied to the model was suitable for analysis and prediction.

The dataset contained variables describing different characteristics of houses, together with the corresponding house price. Numerical variables were prepared for modelling, while the categorical Neighborhood variable was converted into a numerical representation using one-hot encoding.

Data Preparation Decisions

Data preparation step	Decision	Reason
Target variable	House_Price_Million_RWF	This is the value the model is required to predict
House identifier	House ID excluded	It is an identifier rather than a meaningful predictive feature
Categorical variable	Neighborhood encoded	Machine learning models require numerical input
Dataset splitting	80% training / 20% testing	Allows model training and independent evaluation
Random state	42	Makes the train-test split reproducible
Model type	Multiple Linear Regression	Suitable for predicting a continuous numerical target

After preparation, the cleaned dataset was saved as:

house_price_prediction_cleaned.csv

The cleaned dataset was then used as the basis for model development and evaluation.

3. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand the structure of the dataset and investigate relationships between the house characteristics and house prices.

The analysis included visualizations of important numerical variables and their relationship with the target variable. These visualizations helped identify patterns and provided an initial understanding of the factors associated with house prices.

Figure 1: Distribution of House Prices

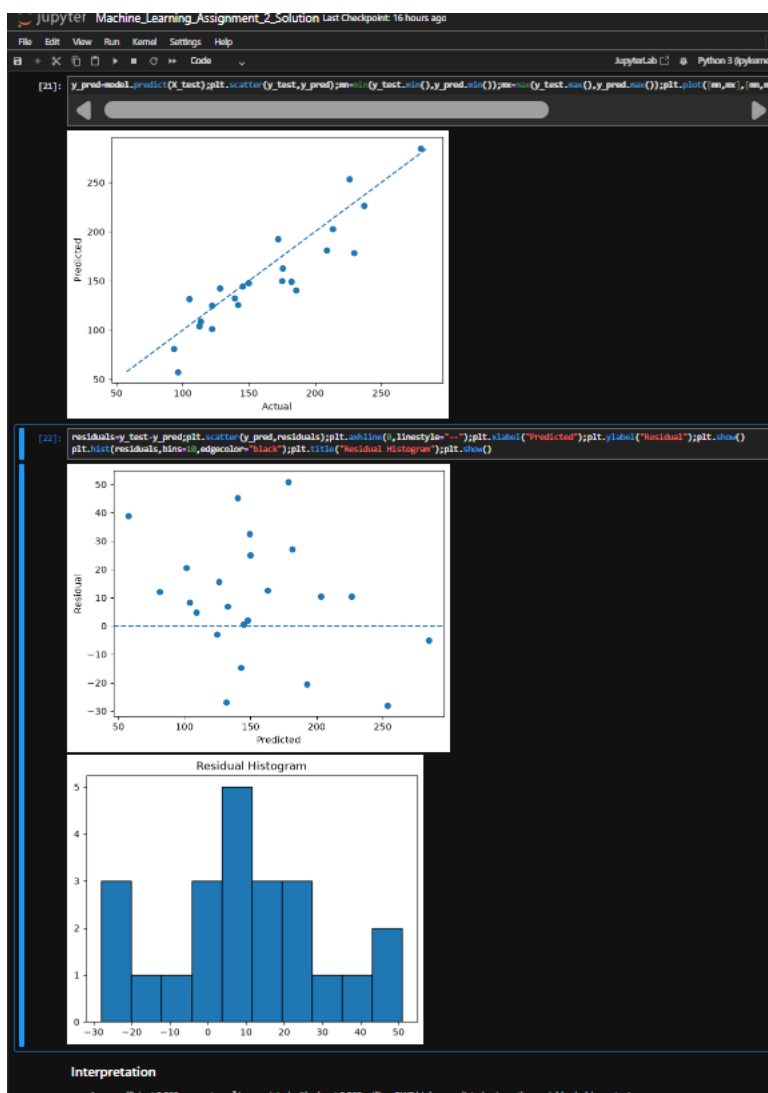


Figure 2: Relationship Between House Area and House Price

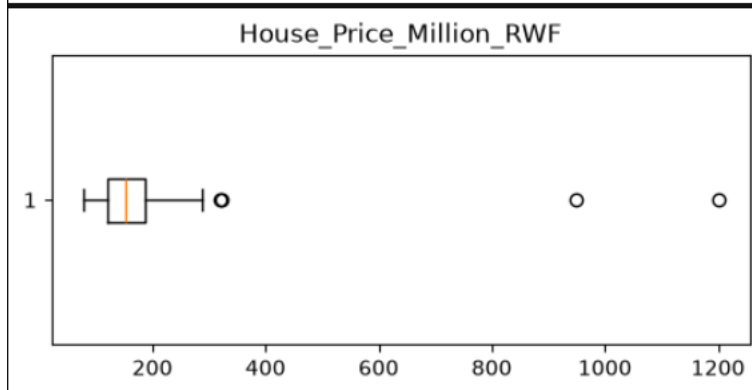
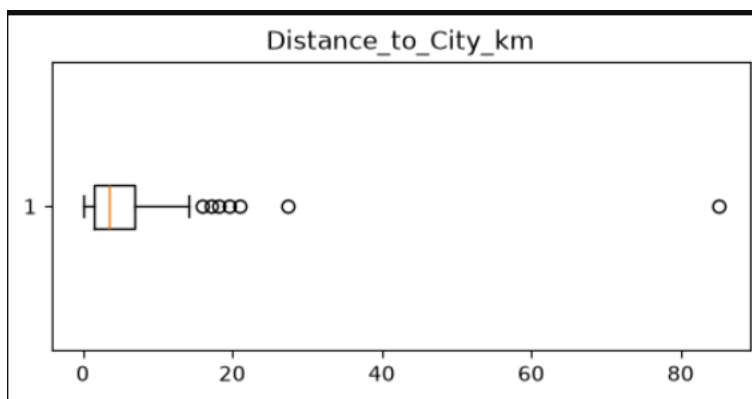
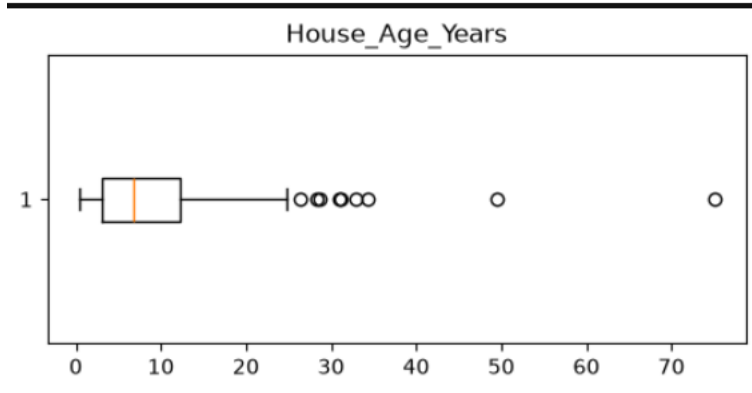
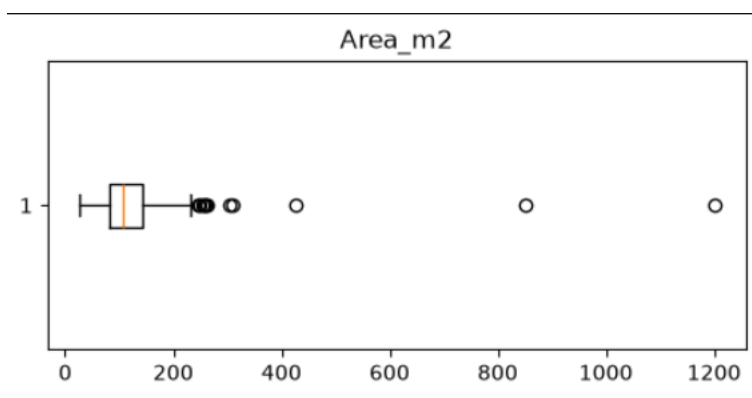
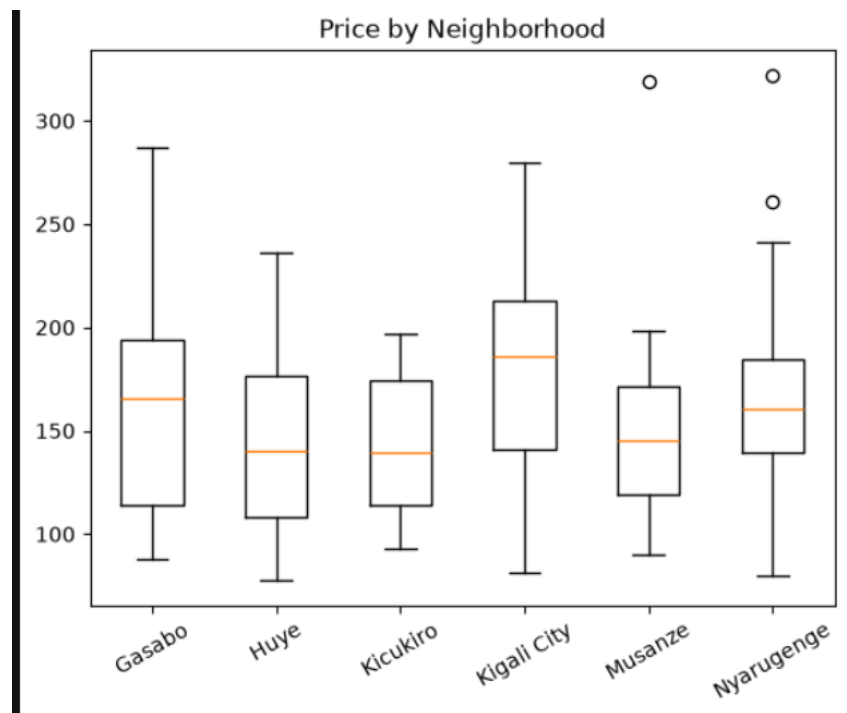
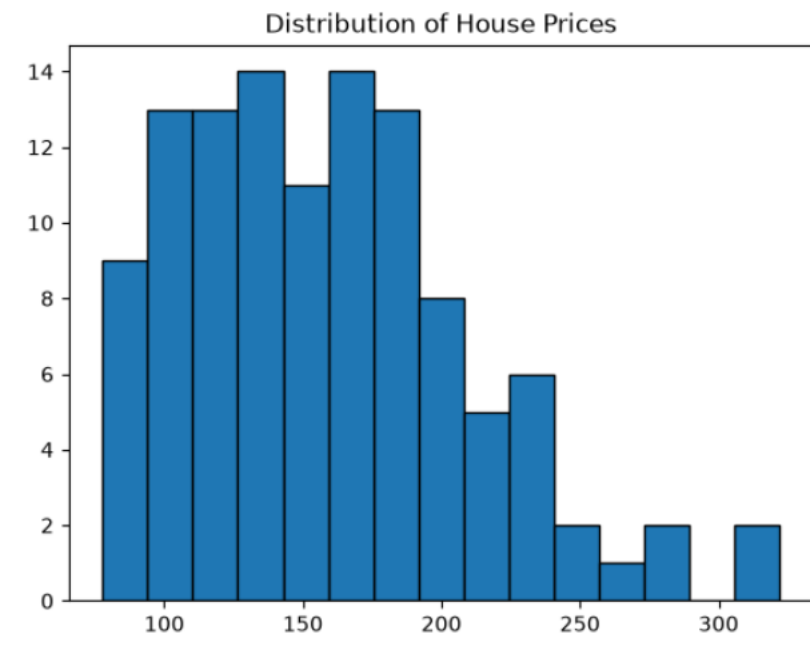


Figure 3: House Price Relationship by Neighborhood or Other Important Feature



The exploratory analysis provided a basis for selecting the variables used in the prediction model and helped in interpreting the model results.

4. Model Development and Results

A Multiple Linear Regression model was developed to predict house prices. Multiple Linear Regression was selected because the target variable, House_Price_Million_RWF, is continuous and the project aims to estimate a numerical house price from several input variables.

The dataset was divided into training and testing sets using an 80/20 split. A random_state value of 42 was used to ensure that the same division could be reproduced.

The Neighborhood categorical variable was processed using one-hot encoding. The model was then trained using the prepared training data.

Model Configuration

Parameter	Value
Algorithm	Multiple Linear Regression
Target	House_Price_Million_RWF
Training data	80%

The model was trained using the prepared training dataset and evaluated using the separate testing dataset. The evaluation was carried out using the coefficient of determination (R^2) and Root Mean Squared Error (RMSE).

Model Evaluation Results

Evaluation Metric	Result
R^2 Score	0.7879
RMSE	22.94 million RWF

The model achieved a test R^2 score of **0.7879**. This means that approximately **78.79% of the variation in house prices in the test dataset is explained by the variables included in the model.**

The model also achieved a Root Mean Squared Error (RMSE) of **22.94 million RWF**. Since the target variable is measured in millions of Rwandan francs, this means that the model's prediction errors have a typical magnitude of approximately 22.94 million RWF according to the RMSE measure.

These results indicate that the selected house characteristics provide useful information for predicting house prices. However, the remaining unexplained variation also shows that additional factors may influence house prices and that there is room for further improvement.

Regression Coefficients

The regression coefficients describe the relationship between each predictor and the predicted house price while the other variables are held constant.

6. Application Deployment

After training and evaluating the model, the trained model was saved as:

house_price_model.sav

A Streamlit web application was developed using app.py. The application allows a user to enter relevant house characteristics and obtain a predicted house price.

The application was first tested locally using Streamlit. The local application was accessed through:

<http://localhost:8501>

Figure 6: Local Streamlit House Price Prediction Application

[Insert screenshot of the local application showing a prediction here.]

The application was subsequently deployed online using Streamlit Community Cloud. The deployed application can be accessed at:

<https://ikyec6bhqdrqvp8spn2hjc.streamlit.app/>

The deployment demonstrates how a trained machine learning model can be integrated into a user-facing application. Instead of requiring users to interact directly with the Python code or Jupyter Notebook, the Streamlit interface allows them to provide property information and receive a house-price prediction.

7. Conclusion

This project demonstrated the complete process of developing and deploying a machine learning solution for house-price prediction. The workflow included data preparation, exploratory data analysis, model development, evaluation, diagnostic analysis, and deployment.

A Multiple Linear Regression model was developed using an 80/20 training and testing split with a random state of 42. The Neighborhood variable was converted into numerical features using

one-hot encoding, while House_ID was excluded because it is an identifier rather than a meaningful predictive variable.

The final model achieved a test R^2 score of **0.7879** and an RMSE of **22.94 million RWF**. The results demonstrate that the selected property characteristics can provide useful information for estimating house prices, while also showing that some variation in house prices remains unexplained.

The project also successfully demonstrated the practical deployment of the trained model through a Streamlit application. The application provides a simple interface for entering house characteristics and obtaining a predicted price.

8. Limitations

The project has several limitations that should be considered when interpreting the results.

First, the model's performance depends on the quality and representativeness of the available dataset. A dataset that does not fully represent the housing market may limit the model's ability to generalize to new properties.

Second, Multiple Linear Regression assumes relationships between the predictors and target that can be reasonably represented using a linear model. Real-world house prices may involve more complex relationships.

Third, the available dataset may not contain all factors that influence house prices. Factors such as the number of bedrooms and bathrooms, land size, property condition, road accessibility, nearby services, security, and detailed geographic location could also affect property prices.

Finally, the reported evaluation results are based on the available test dataset. They should therefore be interpreted as an evaluation of the model on this dataset rather than a guarantee of prediction accuracy for every future property.

Additional features could be created from the existing variables to represent relationships that may be useful for prediction.

9.6 Improve the Streamlit Application

The application could be enhanced with additional validation, interactive visualizations, clearer prediction explanations, and more user-friendly input controls.

9.7 Update the Model with New Data

As new housing-market data becomes available, the model could be retrained periodically to keep its predictions relevant to changing market conditions.

Final Project Files

The final project repository contains the following files:

App.py Streamlit prediction application

house_price_model.sav trained machine learning model

Machine_Learning_Assignment_2_Solution.ipynb Jupyter Notebook containing the machine learning workflow

requirements.txt Python dependencies

house_price_prediction_cleaned.csv cleaned dataset

Part_G_Report.pdf final project report

README.md project documentation

GitHub Repository:

<https://github.com/Wildlife91/HOUSE-PRICE-PREDICTOR-.git>

Live Application: <https://ikyec6bhqdrqvp8spn2hjc.streamlit.app/>